

Markov Decision Processes (MDP)

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1 Markov Decision Processes

A Markov decision process (MDP) is a tuple (S, A, R, P) with S being the states, A being the actions, rewards $R = R(s, a)$ which you get when in state s and applying action a and a transition probability $P = P(s' | s, a)$ which gives the probability of ending in state s' if we are at state s and apply action a . Additional information can be an initial state probability distribution $P(s_0)$, which tells us where we likely will start. If rewards are non-deterministic (same action and state transition might lead to different rewards), you need to replace $R(s, a)$ by a probability distribution $P(r | s, a)$. We consider here only *stationary* MDPs, which are independent of time. To define a *non-stationary* MDP, every state, action and reward becomes time dependent, i.e. $s \rightarrow s_t$, $a \rightarrow a_t$ and $r \rightarrow r_t$.

Given an MDP, our goal is to find an optimal policy $\pi^*(s)$. A policy tells us, in each state, which action to take. An optimal policy gives us the action which will optimize a cost function over rewards. A commonly used cost function is the expected discounted reward over an infinite horizon, which we define as

$$\pi^*(s) = \operatorname{argmax}_{\pi} E_{\pi} \left\{ \sum_{i=0}^{\infty} \gamma^i r_i \mid s \right\}. \quad (1)$$

There are two main motivations for this formulation. First, it makes intuitively sense to maximize the cumulative reward over all future actions. Rewards close in time should have a bigger influence because we should try to avoid disastrous outcomes (airplane crashing, humanoid robot falling down). Actions further away in time could potentially be fixed at a later time and are therefore less important. Second, this formulation allows us to write algorithms which can be shown to converge to the optimal policy. This would, however, not work if we use e.g. $\gamma = 1$.

2 Episodes and terminal states

a) Let us add a Nirvana state to our MDP, i.e. we increase our state space with $S = S \cup s_N$. This state has transition probability

$$P(s' | s = \{s_N, s_T\}, a) = \begin{cases} 1 & \text{if } s' = s_N \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

and returns a reward of $R(s_N, a) = 0$. The total return for the agent is then given by

$$\sum_{t=0}^{\infty} \gamma^t r_t = \sum_{t=0}^T \gamma^t r_t + \sum_{t=T}^{\infty} \gamma^t R(s_N, a) = \sum_{t=0}^T \gamma^t r_t. \quad (3)$$

b) The conceptual idea is to apply the Bellman principle of optimality. Let us assume that we have an optimal policy π_1 for the non-tunnel MDP and an optimal policy π_2 for the tunnel MDP. Are those policies the same?

To see if they are, we first will have a closer look at Bellman's principle of optimality. The optimality principle states that if you have an optimal policy (for some cost function) from, let's say, a state A to a state B , then every sub-policy from a state C to B must constitute an optimal policy with regard to C . To see this, imagine that there would be a better policy from C to B than the sub-policy of the optimal policy. In that case, we could update the optimal policy. However, this contradicts our statement that we had an optimal policy to begin with. Note that the principle only applies to problems which exhibit an optimal substructure, i.e. the cost function is additive.

For our example, the tunnel policy π_2 consists of sub-policies from s_0 to $s_T^{(0)}$, from s_0 to $s_T^{(1)}$ and so on. Every sub-policy is optimal according to the Bellman principle and therefore needs to be equivalent to the non-tunnel policy π_1 . Therefore, the policies must be equivalent. However, the value function will differ by some factor.